

Aayush Sood

aayush.sood@uwaterloo.ca | /in/soodaayush | github.com/soodaayush | aayushsood.com | Canadian Citizen

EDUCATION

University of Waterloo

Honours Software Engineering, Co-op

Waterloo, ON

2026 - 2031

PROJECTS

IWKids - Mobile App | React Native, JavaScript, Expo, Git, OpenAI API ([GitHub](#))

- Engineered a cross-platform **React Native** mobile app integrating the **OpenAI API** to automate pediatric ER workflows and provide real-time conversational support for patients and families
- Placed 2nd out of 24 teams** at the Atlantic Canada AI Summit Youth Pitch Competition, hosted by Saint Mary's University and IWK Hospital, recognized for technical execution, practicality, and high-impact healthcare delivery

SAP-1 - 6502 Microprocessor / 8-bit Project | Python, Assembly ([GitHub: 6502](#)) ([GitHub: 8-bit](#))

- Designed and assembled functional 8-bit and 6502 based breadboard computers from discrete logic chips (65C02 and 74LS- series), implementing the SAP-1 architecture to master hardware-level CPU operations.
- Programmed low-level routines in Assembly to run working programs on the physical hardware, debugged power, memory addressing, and signal issues directly on the breadboard

CoinDeno | React.js, React Native, REST, Git, Firebase, CoinGecko APIs ([GitHub](#)) ([Link](#))

- Built and shipped a crypto portfolio tracker to the **App Store and Google Play**, reaching **20+ users**
- Implemented secure Google OAuth authentication and integrated CoinGecko REST API for live portfolio tracking
- Extended to React Native for cross-platform mobile support with shared state logic across iOS and Android

Market Maker | Python ([GitHub](#))

- Engineered a **Python**-based trading system simulation that dynamically adjusts buy/sell spreads and asset pricing based on real-time market volatility and automated inventory-level logic
- Built a multi-agent testing environment to model diverse user behaviors under fluctuating market conditions, leveraging **Matplotlib**, **NumPy** to generate automated data visualizations tracking profit metrics

EXPERIENCE

Stanford University's Code in Place - Section Lead

May 2025, May 2026

- Selected to lead weekly Python instruction for two cohorts totaling **30+ global learners** as part of Stanford's flagship CS106A global education initiative
- Facilitated interactive technical discussions and code reviews, mentoring students through debugging strategies and core programming concepts (data structures, logic, decomposition) while achieving high satisfaction feedback from students

Nova Scotia Secondary School Students' Association - IT Administrator

Apr 2025 – Jun 2026

- Administered IT infrastructure for **100+ provincial users** across M365, Entra, and Google Workspace, resolving **20+ help desk tickets** and authoring workflow documentation
- Led the website redesign for www.nsssa.ca, improving accessibility and information flow for users

Tech Club President

Sep 2023 – Jun 2026

- Grew student membership **from 3-4 to 30+** and delivered **80+ technology workshops** spanning web development, AI, cybersecurity, and Arduino/embedded programming, sustaining consistent engagement across three academic years
- Organized and managed **5 large-scale events** averaging **40+ attendees** and **~\$5k in budget**

HONORS & AWARDS

Schulich Leader Scholarship (McGill): Winner of Canada's premier \$100k STEM undergraduate scholarship (2026)

Scotiabank Software Engineering Scholarship (UWaterloo): \$20k entrance scholarship for outstanding students (2026)

Wolf Scholar Finalist (UofT): Named a national finalist for Inaugural Wolf Scholarship (1 of 32 selected nationwide) (2026)

AFA CyberPatriot: Won Gold Tier in an international, team-based cyber defense competition (2025)

IWK AI Pitch Competition: Won 2nd Place for designing an AI-driven pediatric healthcare workflow solution (2025)

Lockheed Martin CYBERQUEST Competition: Won 3rd Place in a team-based, global cybersecurity CTF challenge (2025)

Canadian Computing Competition Senior Division: Certificate of Distinction, Score: 45/75 (2025)

Canadian Computing Competition Junior Division: Certificate of Distinction & Honour Rolls, Score: 66/75 (2024)

Skills

Languages: Python, C++, JavaScript, SQL

Frameworks: React, React Native, Node.js, HTML, CSS/SCSS, Tailwind

Tools: Firebase, Cloudflare, Netlify, Git, GitHub, MS Azure, Linux

Hardware: 6502 Microprocessor, 8-Bit Breadboard CPU, Arduino